Exam AWS Certified Developer - Associate DVA-C02 All Questions

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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 7
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A developer has an application that makes batch requests directly to Amazon DynamoDB by using the BatchGetItem low-level API operation. The responses frequently return values in the UnprocessedKeys element.
Which actions should the developer take to increase the resiliency of the application when the batch response includes values in UnprocessedKeys? (Choose two.)

A. Retry the batch operation immediately.

B. Retry the batch operation with exponential backoff and randomized delay.

C. Update the application to use an AWS software development kit (AWS SDK) to make the requests.

D. Increase the provisioned read capacity of the DynamoDB tables that the operation accesses.

E. Increase the provisioned write capacity of the DynamoDB tables that the operation accesses.

Show Suggested Answer

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by aragon_saa at *March 16, 2023, 10 a.m.*

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Comments

mrsoa Highly Voted 2 years, 7 months ago

Selected Answer: BD

B D

From Stephan's maarek course

BatchGetItem
• Return items from one or more tables
• Up to 100 items, up to 16 MB of data
• Items are retrieved in parallel to minimize latency
• UnprocessedKeys for failed read operations (exponential backoff or add RCU)

upvoted 9 times

aanataliya Highly Voted 2 years, 7 months ago

Selected Answer: BD

B and D is correct answer. AWS SDK automatically takes care of both retry and exponential backoff. If we choose C, selecting only C will answer our question(no need of B) but We need to choose 2 answer. In addition, question doesnot specifically say to change core logic from low level api to SDK. by choosing B and D we can improve resiliency. https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Programming.Errors.html#Programming.Errors.RetryAndBackoff

upvoted 6 times

lebbar Most Recent 1 week, 1 day ago

Selected Answer: BD

B D
When DynamoDB returns UnprocessedKeys, it means it could not process some keys from your batch request. Two possible reasons:

The table has insufficient provisioned read capacity (throttling)
The request exceeded DynamoDB limits (16 MB max, 100 items max per batch)

upvoted 1 times

wtf3344 1 year, 2 months ago

Selected Answer: BD

BatchWriteItem
• UnprocessedItems for failed write operations (exponential backoff or add WCU)
BatchGetItem
• UnprocessedKeys for failed read operations (exponential backoff or add RCU)
So answer is B & D

upvoted 2 times

Arad 1 year, 3 months ago

Selected Answer: BC

BC is correct answer.

upvoted 2 times

sumanshu 1 year, 3 months ago

Selected Answer: BD

DynamoDB BatchGetItem API operation, the request fetches multiple items in one operation. However, DynamoDB has limits on the resources it allocates for processing requests. If it cannot process some items within a batch request due to these limits, it returns the processed items (the ones it successfully retrieved), and for the remaining items, it includes them in the UnprocessedKeys element in the response.

upvoted 2 times

trieudo 1 year, 3 months ago

Selected Answer: BC

Keyword: increase, resiliency, application, response, UnprocessedKeys(Unprocessed due to be not enough infra)

==> Discard A: horrible way, push so many traffic to busy system
==> Discard D, E: Error due to huge traffic, it can be unlimited. Don't come from lacking resource ==> Your resource, your money is limit >< unpredicted huge traffic

B: is good way to call x2 times after failed, call again after 1s -> 2s -> 4s -> 8s -> ... until success

C: is intergrate method of 'B' option automaically inside SDK

upvoted 1 times

9d8dd9c 1 year, 5 months ago

Selected Answer: BD

BD
C handles retry but using SDK is not necessary here

upvoted 1 times

eberhe900 1 year, 6 months ago

Selected Answer: BD

Backoff and Retry Strategy:

Implement a backoff and retry strategy to prevent overwhelming the DynamoDB service with repeated requests.
Apply an exponential backoff algorithm, where you progressively increase the delay between each retry attempt.
Consider implementing a maximum number of retries to avoid an infinite retry loop.
Fine-Tuning DynamoDB Provisioned Capacity:

If you consistently encounter unprocessed items during batch operations, it may indicate that your DynamoDB table's provisioned capacity is insufficient.
Monitor the table's consumed capacity and adjust the provisioned capacity (read capacity units) accordingly to handle the load and reduce the occurrence of unprocessed items.
By implementing these steps, you can effectively handle unprocessed items returned by the BatchGetItem operation in DynamoDB and ensure that all items are processed successfully.

upvoted 2 times

SherzodBek 1 year, 6 months ago

Selected Answer: BD

B & D.
B is correct. Because in the question, it is mentioned that low-level API is being used.It means exponential backoff can be implemented manually.
D is correct. Because there is a frequently keyword in the question. If UnprocessedKeys error occurs frequently, DynamoDB doesn't have enough capacity to process requests. So read capacity should be increased.

upvoted 3 times

TheFivePips 1 year, 6 months ago

Selected Answer: BD

Exponential backoff with randomized delay is a common technique used to handle transient failures and throttle errors in distributed systems like DynamoDB. This approach involves retrying the failed operation after waiting for an increasing amount of time, which helps reduce the load on the service and increases the likelihood of success during periods of high demand or throttling.

If the BatchGetItem operation frequently returns values in the UnprocessedKeys element, it indicates that the table's read capacity might be insufficient to handle the requested workload. By increasing the provisioned read capacity for the DynamoDB tables, the application can better handle the read throughput requirements and reduce the likelihood of encountering UnprocessedKeys in batch responses.

AWS SDK might provide additional features and simplifications for making requests, it does not directly address the issue of UnprocessedKeys in batch responses. This option might be beneficial for improving code maintainability and leveraging SDK features however.

upvoted 1 times

65703c1 1 year, 10 months ago

Selected Answer: BC

BC is the correct answer.

upvoted 1 times

drycleansing 1 year, 12 months ago

Selected Answer: BC

among B C D, it is hard to say D copes with the problem directly I guess. Increasing RCU will affects the ratio of unprocessed items but that does not mean it handles the unprocessed items.

upvoted 1 times

vinfo 1 year, 12 months ago

Selected Answer: BD

B,D. La combinación de estrategias es ideal para este comportamiento

upvoted 1 times

apa_1 2 years ago

Selected Answer: BD

B,D is correct

upvoted 1 times

DeaconStJohn 2 years ago

Selected Answer: BC

the aws documentation for unprocessedkeys reads:
"If DynamoDB returns any unprocessed items, you should retry the batch operation on those items. However, we strongly recommend that you use an exponential backoff algorithm. If you retry the batch operation immediately, the underlying read or write requests can still fail due to throttling on the individual tables. If you delay the batch operation using exponential backoff, the individual requests in the batch are much more likely to succeed."

Therefore I am taking the two options that provide this functionality.

https://docs.aws.amazon.com/amazondynamodb/latest/APIReference/API_BatchGetItem.html

upvoted 2 times

ibratoev 2 years ago

Selected Answer: BD

B & D is correct

upvoted 1 times